Linear Algebra I – Week 7

Independent Study: Proposed Problems (50)

Fall 2026

Topics: transposes and permutations; vector spaces and subspaces.
Reference: G. Strang, *Introduction to Linear Algebra*, 4th ed., §2.7 (pp. 108–120) and §3.1 (pp. 121–132).
Instructions: Use column-vector notation, show intermediate steps, and justify subspace answers. Problems are ungraded independent practice; no solutions are included.

Part A: Transposes and elementary operations

Exercise 1. Compute A^T for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Exercise 2. Compute $(A+B)^T$ for $A = \begin{pmatrix} 1 & 0 \\ 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 1 \\ 0 & -1 \end{pmatrix}$.

Exercise 3. Find A^TA for $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$.

Exercise 4. Find $(AB)^T$ for $A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$.

Exercise 5. Is $A = \begin{pmatrix} 2 & -3 \\ -3 & 1 \end{pmatrix}$ symmetric?

Exercise 6. Decompose $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$ into symmetric and skew parts.

Exercise 7. Find PA where $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Exercise 8. Find AP for the same A and P.

Exercise 9. Find P^{-1} for $P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$.

Exercise 10. Find P exchanging rows 1 and 3 of a 3×3 matrix.

Exercise 11. Find P, L, U for $A = \begin{pmatrix} 0 & 2 \\ 3 & 4 \end{pmatrix}$.

Exercise 12. Find L, U for $A = \begin{pmatrix} 2 & 1 \\ 6 & 5 \end{pmatrix}$ without row swaps.

Part B: Permutations and pivoting

Exercise 13. Is $W = \{(x, y) : y = 3x\}$ a subspace of $\mathbb{R}^2$?

Exercise 14. Is $S = \{(x, y) : y = 3x + 2\}$ a subspace?

Exercise 15. Is $W = \{(x, y, z) : x + y - z = 0\}$ a subspace?

Exercise 16. Express $(4, 1)$ as a combination of $(1, 1)$ and $(1, -1)$.

Exercise 17. Find the span of $(1, 2, 0)$ and $(2, 4, 0)$.

Exercise 18. Find Null(A) for $A = \begin{pmatrix} 1 & 2 & -1 \end{pmatrix}$.

Exercise 19. Find Col(A) for $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 0 \end{pmatrix}$; is $(3, 6, 0)$ in it?

Exercise 20. For $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ find A^T and $\det(A)$.

Part C: Subspace criteria

Exercise 21. For $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$, compute A^T and identify dimensions.

Exercise 22. Prove $(A + B)^T = A^T + B^T$ using entries.

Exercise 23. Prove $(AB)^T = B^T A^T$ for compatible matrices.

Exercise 24. Compute $A^T A$ for $A = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 3 \end{pmatrix}$.

Exercise 25. For $A = \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix}$, find symmetric and skew-symmetric parts.

Exercise 26. Show that $A^T A$ is symmetric for every real matrix A.

Exercise 27. Compute $u^T v$ and uv^T for $u = (2, -1)^T$, $v = (3, 4)^T$.

Exercise 28. Construct the permutation matrix exchanging rows 1 and 2 of I_3.

Exercise 29. For $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, compare PA and AP when $A = \begin{pmatrix} 2 & 5 \\ 7 & 1 \end{pmatrix}$.

Exercise 30. For $P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$, verify $P^T P = I$.

Exercise 31. Find the determinant of the permutation matrix that exchanges rows 1 and 3.

Exercise 32. Find P, L, U for $A = \begin{pmatrix} 0 & 1 \\ 4 & 2 \end{pmatrix}$.

Part D: Spans and matrix subspaces

Exercise 33. Use partial pivoting for $A = \begin{pmatrix} 0.001 & 1 \\ 2 & 1 \end{pmatrix}$; compare the multipliers before and after swapping.

Exercise 34. Test whether $W = \{(x, y) : x + 2y = 0\}$ is a subspace.

Exercise 35. Test whether $W = \{(x, y) : x + 2y = 3\}$ is a subspace.

Exercise 36. Test whether $W = \{(x, y) : x \geq 0\}$ is a subspace.

Exercise 37. Test whether $W = \{(x, y, z) : x - y + z = 0\}$ is a subspace.

Exercise 38. Express $(6, 2)$ as a combination of $(1, 1)$ and $(1, -1)$.

Exercise 39. Describe $\text{span}\{(1, 2, 0), (0, 1, 2)\}$ geometrically.

Exercise 40. Find $\text{Null}(A)$ for $A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$.

Exercise 41. Find $\text{Col}(A)$ for $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{pmatrix}$.

Exercise 42. Is $(1, 1, 1)$ in $\text{Col}(A)$ for $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$?

Part E: Mixed applications

Exercise 43. Prove that the intersection of two subspaces is a subspace.

Exercise 44. Give two subspaces whose union is not a subspace.

Exercise 45. Determine whether the set of symmetric 3×3 matrices is a subspace of all real 3×3 matrices.

Exercise 46. Determine whether the polynomials satisfying $p(0) = 1$ form a subspace.

Part F: Challenges

Exercise 47. Challenge: Prove that $\text{Null}(A^T A) = \text{Null}(A)$ for every real matrix A.

Exercise 48. Challenge: If P is a permutation matrix, prove $\text{Col}(PA) = P\,\text{Col}(A)$.

Exercise 49. Challenge: If W is a subspace, prove $\text{span}(W) = W$.

Exercise 50. Challenge: For $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, describe $\text{Col}(A)$ and $\text{Null}(A)$.